



HW 4.1: In this question, we will explore the relativistic correction to the energy levels of the hydrogen atom. This is called the *fine structure correction*. Knowing that the energy of a relativistic particle is given by $E = \sqrt{p^2 c^2 + m^2 c^4}$, we can expand this expression in powers of p/mc to find the leading order relativistic correction to the energy of the electron.

- a) Write down the Hamiltonian with the leading relativistic correction to the hydrogen atom.
- b) Argue why we are justified to use non-degenerate perturbation theory to calculate the effects of this correction on the energy levels of the hydrogen atom, even though there are degeneracies in the spectrum of $\hat{H}_0$. Are the eigenstates of $\hat{H}_0$ a “good basis” for our perturbation?
- c) Using the unperturbed Schrödinger equation,

$$\hat{\mathbf{p}}^2 |n, l, m\rangle = 2m(E_n^{(0)} - V(r)) |n, l, m\rangle, \quad (1)$$

show that the leading order relativistic correction to the energy levels of the hydrogen atom can be written as

$$\Delta E_{\text{rel}} = -\frac{1}{2mc^2} \langle n, l, m | ((E_n^{(0)})^2 - 2E_n^{(0)}V(r) + V(r)^2) | n, l, m \rangle. \quad (2)$$

- d) Show that the first order correction is given by

$$\Delta E_{\text{rel}} = -\frac{(E_n^{(0)})^2}{2mc^2} \left(\frac{4n}{l + 1/2} - 3 \right), \quad (3)$$

where E_n is the non-relativistic energy of the level, n is the principal quantum number, and l is the orbital angular momentum quantum number.

- e) What order in $\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c}$ is the relativistic correction? What order do you expect the next relativistic corrections to be? Why?

- f) The degeneracy of the energy levels of the hydrogen atom in ℓ was explained by the conservation of a vector called the Runge-Lenz vector:

$$\hat{\mathbf{A}} = \frac{1}{2m}(\hat{\mathbf{p}} \times \hat{\mathbf{L}} - \hat{\mathbf{L}} \times \hat{\mathbf{p}}) - k\frac{\hat{\mathbf{r}}}{r}, \quad (4)$$

where k is the positive constant in our *central* potential. With the Heisenberg picture in mind, show that the relativistic correction breaks the conservation of the Runge-Lenz vector.

(Hint: use the fact that $[\hat{H}_0, \hat{\mathbf{A}}] = 0$. You also may need $[\mathbf{V}(\mathbf{r}), \hat{\mathbf{p}}] = -i\hbar\nabla V(r)$ and $[\hat{\mathbf{r}}, \hat{\mathbf{L}}] \neq 0$).